

Ticket System Final Project

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Overview

- This program is a miniature ticket master type system
- In it there are venues that hold events, guests who purchase tickets to these events, and the ticket manager system to orchestrate it
- This system allows you to act on behalf of guests, and venues, but doesn't have true user log-ins. The imagined user of this program would be an admin user.

Program Flow

- The GUI for this program is the console
- The user is presented with menus that have numbered options they can choose from
- The main options are adding either guests or venues, acting on behalf of guests or venues, and admin options
- When acting on behalf of venues you can create events
- When acting on behalf of guests you can purchase tickets

Classes

- TicketMgr
 - Orchestrates all actions taken
 - Contains registries of the other classes and is responsible for saving to and loading from the file
- Guest
 - Guest name and list of purchased tickets
- Venue
 - Name and list of events

Classes cont.

- Event
 - Title
 - Ticket information
 - Types (Standard and VIP)
 - Seats available (purchased with VIP tickets)
 - Pricing
 - List of sold tickets
- Ticket
 - Pointer to event and guest
 - Purchase price

Data Structures

- The primary data structure used in this program is vectors of pointers. For example, GuestList is a vector of guest pointers
- There are specialized data structures for loading data from a json file
 - The classes have circular references, so data must be loaded into a temporary structure, so all connections can be made
- Additionally, unordered maps with specialized structs are used in various cases
 - E.g. To display the events that a guest is attending, a map is created to group tickets by event and print out this consolidated info

Thank
you